随机过程

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Finite-state Markov Chains

Review

§ Eigenvalues and Eigenvectors of Stochastic Matrices

§ Markov chains with rewards

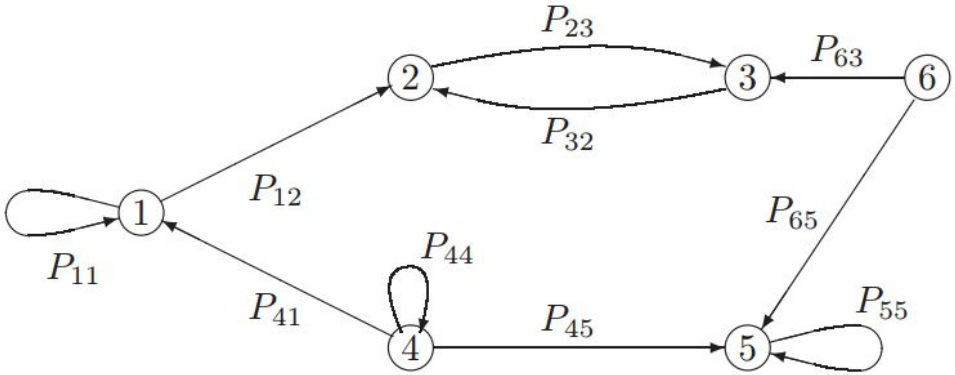
§ Exercise

Definitions of Markov Chains

Definition 1

A Markov chain is an integer-time process, $\{X_n, n \geqslant 0\}$, for which the sample values for each r.v. $X_n, n \geqslant 1$, lie in a countable set S and depend on the past only through the most recent r.v. X_{n-1}.

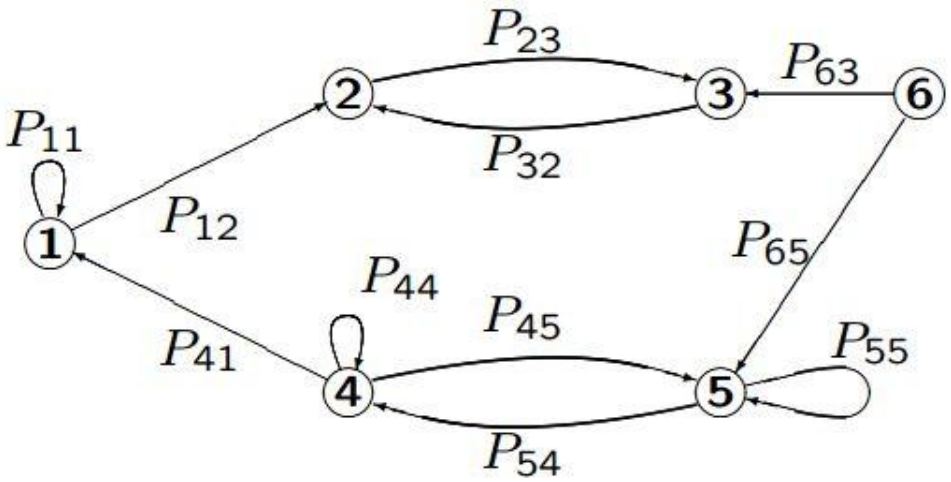
$$\Pr\{X_n | X_{n-1}, \ldots, X_0\} = \Pr\{X_n | X_{n-1}\}.$$



$$[P] = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{16} \\ P_{21} & P_{22} & \cdots & P_{26} \\ \vdots & \vdots & \vdots\vdots\vdots & \vdots \\ P_{61} & P_{62} & \cdots & P_{66} \end{bmatrix}$$

Relative Concepts

- walks, paths, cycles;
- states which are accessible, states which communicate;
- classes;
- recurrent states/classes, transient states/classes;
- periodic states/classes, aperiodic states/classes;
- ergodic classes, ergodic chains;
- unichains, ergodic unichains;



$\mathcal{C}_1 = \{2, 3\}$
$\mathcal{C}_2 = \{4, 5\}$
$\mathcal{C}_3 = \{1\}$
$\mathcal{C}_4 = \{6\}$

Classification of States

Theorem 1

For finite-state Markov chains, either all states in a class are transient or all are recurrent.

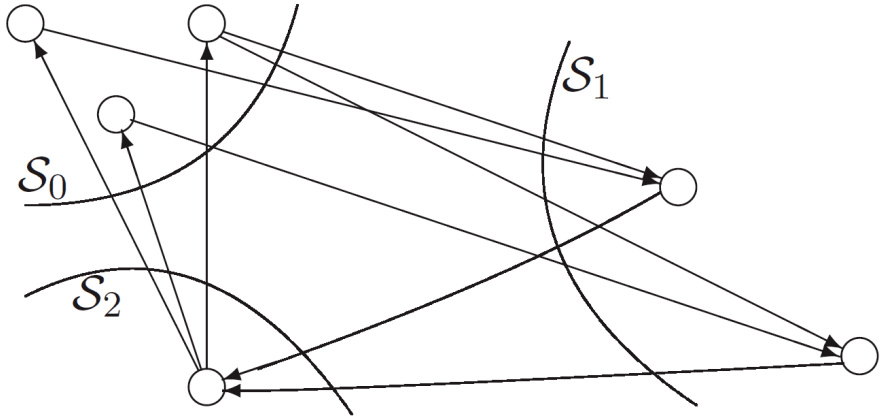
Theorem 2

For any Markov chain (with either a finite or countably-infinite number of states), all states in the same class have the same period.

Classification of States

Theorem 3

If a class C in a finite-state Markov chain has period d, then the states in C can be partitioned into d subsets, $S_0, S_1, \ldots, S_{d-1}$, in such a way that all transitions from S_ℓ go to $S_{\ell+1}$ for $\ell < d-1$, and go to S_0 for $\ell = d-1$.



For a unichain with transition matrix $[P]$, the d-step transition (with transition matrix $[P^d]$) has exactly d recurrent classes.

Classification of States

Theorem 4

For an ergodic M state Markov chain, $P_{ij}^m > 0$ for all i, j, and all $m \geqslant (M-1)^2 + 1$.

Definition 2

A steady-state vector (or a steady-state distribution) for an M-state Markov chain with transition matrix $[P]$ is a row vector $\boldsymbol{\pi}$ that satisfies

$$\boldsymbol{\pi} = \boldsymbol{\pi}[P],$$

where $\sum_i \pi_i = 1$ and $\pi_i \geqslant 0$ for $1 \leqslant i \leqslant M$.

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Eigenvalues and Eigenvectors of Stochastic Matrices

A matrix $[A]$ is singular $\Leftrightarrow$ there is a vector $\boldsymbol{v} \neq 0$ such that $[A]\boldsymbol{v} = 0$;
$\Leftrightarrow$ the columns of $[A]$ are linearly dependent;
$\Leftrightarrow$ the rows of $[A]$ are linearly dependent;
$\Leftrightarrow$ the determinant $\det[A]$ of $[A]$ is 0.

Thus λ is an eigenvalue of $[P]$ if and only if $[P - \lambda I]$ is singular.

Definition 3

A row vector $\boldsymbol{\pi}$ is a left eigenvector of $[P]$ of eigenvalue λ if $\boldsymbol{\pi} \neq 0$ and $\boldsymbol{\pi}[P] = \lambda\boldsymbol{\pi}$, i.e.,$\sum_i \pi_i P_{ij} = \lambda\pi_j$ for all j.
A column vector $\boldsymbol{\nu}$ is a right eigenvector of $[P]$ of eigenvalue λ if $\boldsymbol{\nu} \neq 0$ and $[P]\boldsymbol{\nu} = \lambda\boldsymbol{\nu}$, i.e.,$\sum_j P_{ij}\nu_j = \lambda\nu_j$ for all i.

For every stochastic matrix $[P]$, $[P]\boldsymbol{e} = \boldsymbol{e}$ and thus $[P - I]$ is singular and there is an eigenvalue $\lambda = 1$.

Eigenvalues and eigenvectors for $M = 2$ states

The left and right eigenvector equations can be written out as

$$\begin{cases} \pi_1 P_{11} + \pi_2 P_{21} = \lambda \pi_1 \\ \pi_1 P_{12} + \pi_2 P_{22} = \lambda \pi_2 \end{cases}, \qquad \begin{cases} P_{11}\nu_1 + P_{12}\nu_2 = \lambda \nu_1 \\ P_{21}\nu_1 + P_{22}\nu_2 = \lambda \nu_2 \end{cases}.$$

Each set of equations has a non-zero solution if and only if the matrix $[P - \lambda I]$ is singular, i.e.,

$$(P_{11} - \lambda)(P_{22} - \lambda) - P_{12}P_{21} = 0.$$

$$\lambda_1 = 1, \qquad \lambda_2 = 1 - P_{12} - P_{21}.$$

- If $P_{12} = P_{21} = 0$ (the chain has 2 recurrent classes), then $\lambda = 1$ has multiplicity 2. Otherwise $\lambda = 1$ has multiplicity 1.
- If $P_{12} = P_{21} = 1$ (the chain is periodic), then $\lambda_2 = -1$.
- Otherwise $|\lambda_2| < 1$.

Eigenvalues and eigenvectors for $M > 2$ states

Case with M distinct eigenvalues:

- Suppose $\lambda_1, \ldots, \lambda_M$ are all distinct. The matrix $[P - \lambda_i I]$ is singular for each i, so there must be a right eigenvector $\boldsymbol{\nu}^{(i)}$ and a left eigenvector $\boldsymbol{\pi}^{(i)}$ for each eigenvalue λ_i.
- The right eigenvectors span M-dimensional space and thus the matrix $[U]$ with column $(\boldsymbol{\nu}^{(1)}, \ldots, \boldsymbol{\nu}^{(M)})$ is non-singular.
- For $i \neq j$,
$$(\lambda_i - \lambda_j)\boldsymbol{\nu}^{(i)} \cdot \boldsymbol{\pi}^{(j)} = \boldsymbol{\nu}^{(i)}[P]\boldsymbol{\pi}^{(j)} - \boldsymbol{\nu}^{(i)}[P]\boldsymbol{\pi}^{(j)} = 0.$$
Thus, if we normalize the left eigenvectors to satisfy $\boldsymbol{\nu}^{(i)} \cdot \boldsymbol{\pi}^{(i)} = 1$ for each i, it turn out to be the rows of $[U^{-1}]$.

Eigenvalues and eigenvectors for $M > 2$ states

We can express $[P^n]$ as

$$[P^n] = [U][\Lambda^n][U^{-1}].$$

where $[\Lambda]$ is the diagonal matrix with entries $\lambda_1, \ldots, \lambda_M$.

Break $[\Lambda^n]$ into M terms,

$$[\Lambda^n] = \sum_{i=1}^{M} [\Lambda_i^n]$$

where $[\Lambda_i^n]$ has λ_i^n in position (i, i) and has zeros elsewhere. Then

$$[P^n] = \sum_{i=1}^{M} \lambda_i^n \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}.$$

Eigenvalues and eigenvectors for $M > 2$ states

Theorem 1

The transition matrix $[P]$ has a left eigenvector $\boldsymbol{\pi}$ of eigenvalue $\lambda_1 = 1$ which is a steady-state vector if it is normalized to $\boldsymbol{\pi} \cdot \boldsymbol{e} = 1$, where $\boldsymbol{e} = (1, \ldots, 1)^T$.

Proof: $[P]$ is a stochastic matrix, $[P]\boldsymbol{e} = \boldsymbol{e}$. Therefore, There must be a left eigenvector $\boldsymbol{\pi}$ for eigenvalue 1. For every j, $1 \leqslant j \leqslant M$,

$$\pi_j = \sum_{k=1}^{M} \pi_k P_{kj}.$$

Taking magnitudes,

$$|\pi_j| \leqslant \sum_{k=1}^{M} |\pi_k| P_{kj}.$$

with equality iff. $\pi_j = |\pi_j| \mathrm{e}^{i\phi}$ for all j and some ϕ.

Eigenvalues and eigenvectors for $M > 2$ states

Summing over j,

$$\sum_{j=1}^{M} |\pi_j| \leqslant \sum_{j=1}^{M} \sum_{k=1}^{M} |\pi_k| P_{kj} = \sum_{k=1}^{M} |\pi_k|.$$

This is satisfied with equality.
Thus, $(|\pi_1|, \ldots, |\pi_M|)$ is a nonnegative vector satisfying the steady-state vector equation. Normalizing to $\sum_j |\pi_j| = 1$, we have a steady-state vector.

Theorem 2

Every eigenvalue λ_k of a stochastic matrix $[P]$ satisfies $|\lambda_k| \leqslant 1$.

Remark: These two theorems are valid for all finite-state Markov chains.

Eigenvalues and eigenvectors for $M > 2$ states

Thus, For the case with M distinct eigenvalues, $[P^n]$ can be written as

$$[P^n] = \boldsymbol{e\pi} + \sum_{i=2}^{M} \lambda_i^n \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}$$

- For ergodic unichains, $\lim_{n\to\infty}[P^n] = \boldsymbol{e\pi}$. This implies that $|\lambda_i| < 1$ for $i = 2, \ldots, M$. In this case, we see that the rate at which $[P^n]$ approaches steady state is given by the second largest eigenvalue in magnitude.
- If a recurrent chain is periodic with period d, it turns out that there are d eigenvalues of magnitude 1, and these are uniformly spaced around the unit circle in the complex plane and $[P^n]$ does not converge.

Eigenvalues and eigenvectors for $M > 2$ states

Case with repeated eigenvalues and M independent eigenvectors: Since the number of linearly-independent eigenvectors of each distinct eigenvalue is equal to the multiplicity of that eigenvalue. We can still choose the left eigenvectors of a given eigenvalue to be orthonormal to the right eigenvectors of that eigenvalue.

After doing this and defining $[U]$ as the matrix with columns $\boldsymbol{\nu}^{(1)}, \ldots, \boldsymbol{\nu}^{(M)}$, we see $[U]$ is invertible and that $[U^{-1}]$ is the matrix with rows $\boldsymbol{\pi}^{(1)}, \ldots, \boldsymbol{\pi}^{(M)}$. We then again have

$$[P^n] = \sum_{i=1}^{M} \lambda_i^n \boldsymbol{\nu}^{(i)} \boldsymbol{\pi}^{(i)}.$$

Eigenvalues and eigenvectors for $M > 2$ states

Example

Consider a Markov chain consisting of $\kappa > 1$ ergodic sets of states.

It is easy to see that there are κ different steady-state vectors. Each ergodic set will have an eigenvalue equal to 1 with a right eigenvector equal to 1 on the states of that set and 0 elsewhere. Then the steady-state vector for each recurrent class is strictly positive for each state of the same recurrent class and is zero for all other states.

Then $[P^n]$ will converge to a block diagonal matrix where for each ergodic set, the rows within that set are the same.

Eigenvalues and eigenvectors for $M > 2$ states

The Jordan form case: There are cases in which one or more eigenvalues of $[P]$ are repeated but the number of linearly independent right eigenvectors for a given repeated eigenvalue is less than the multiplicity of that eigenvalue. Thus $[P]$ cannot be expressed as $[U][\Lambda][U^{-1}]$, where Λ is the diagonal matrix of the eigenvalues, repeated according to their multiplicity. Linear algebra says that there exist an invertible matrix $[U]$ and a Jordan form such as

$$[P] = [U][J][U^{-1}].$$

where [J] is a block diagonal matrix with Jordan blocks $[J_s]$ with respect to the eigenvalue λ_i.

$$[J] = \begin{bmatrix} [J_1] & & \\ & \ddots & \\ & & [J_l] \end{bmatrix}, \qquad [J_s] = \begin{bmatrix} \lambda_s & 1 & & \\ & \lambda_s & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_s \end{bmatrix}.$$

Eigenvalues and eigenvectors for $M > 2$ states

Theorem 3

The transition matrix of a finite-state unichain has a single eigenvalue $\lambda = 1$ with an accompanying left eigenvector $\boldsymbol{\pi}$ satisfying $\boldsymbol{\pi}[P] = \boldsymbol{\pi}$ and a right eigenvector $\boldsymbol{e} = (1, \ldots, 1)^T$.

The other eigenvalues λ_i all satisfy $|\lambda_i| \leqslant 1$.

The inequality is strict unless the unichain is periodic, say with period d, and then there are d eigenvalues of magnitude 1 spaced equally around the unit circle.

If the unichain is ergodic, then $[P^n]$ converges to steady state $\boldsymbol{e\pi}$ with an error in each component that goes to 0 at least as $n^k \lambda_s^n$, where λ_s is the eigenvalue of largest magnitude less than 1 and k is the size of the largest Jordan block for the eigenvalue λ_s.

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Review

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§ Exercise

Markov chains with rewards

Suppose that each state i of a Markov chain is associated with a given reward, r_i.
Letting the rv. X_n be the state at time n, the reward at time n is the rv. $R(X_n)$ that maps $X_n = i$ into r_i for each i.
We will be interested only in expected rewards, so that, for example,the expected reward at time n, given that $X_0 = i$ is

$$\mathrm{E}[R(X_n)|X_0 = i] = \sum_j P_{ij}^n r_j.$$

The expected aggregate reward over the n steps from m to $m+n-1$, conditional on $X_m = i$ is

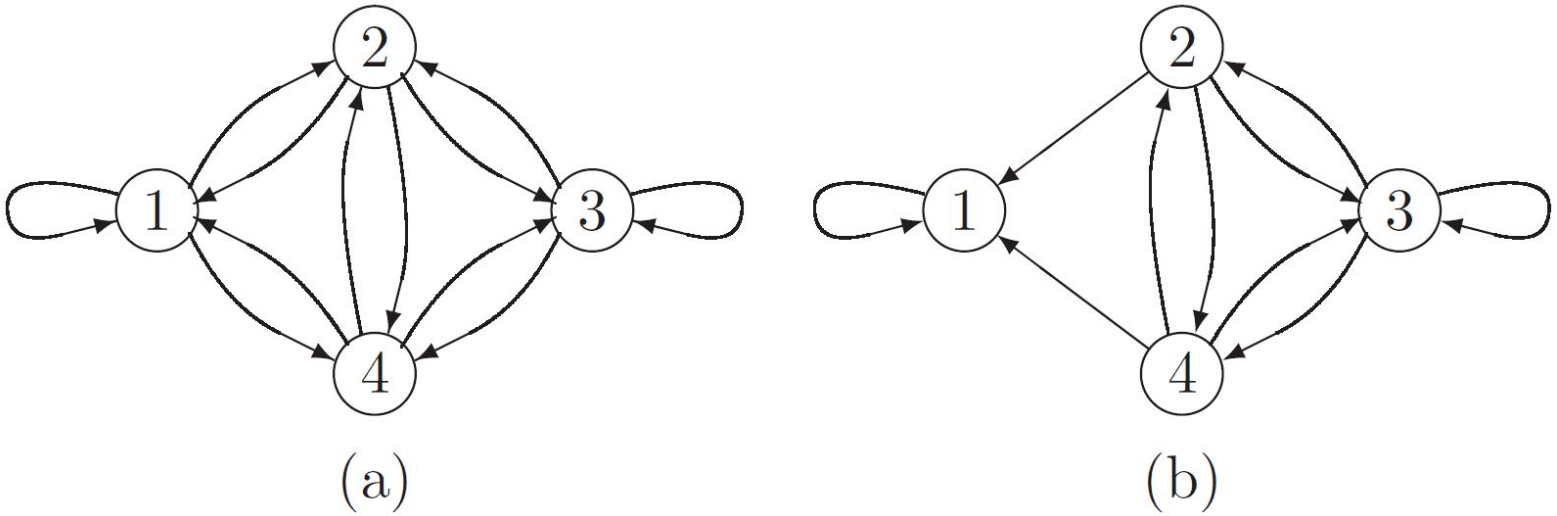
$$\begin{aligned} v_i(n) &= \mathrm{E}[R(X_m) + \cdots + R(X_{m+n-1})|X_m = i] \\ &= r_i + \sum_j P_{ij} r_j + \cdots + \sum_j P_{ij}^{n-1} r_j. \end{aligned}$$

Expected First-passage Times

Suppose, for some arbitrary unichain, that we want to find the expected number of steps, starting from a given state i until some given recurrent state, say state 1, is first entered. Assume $i \neq 1$.

This can be viewed as a reward problem by assigning one unit of reward to each successive state until state 1 is entered.

Thus we can modify the chain to convert the final state 1 into a trapping state: $P_{11} = 1$.



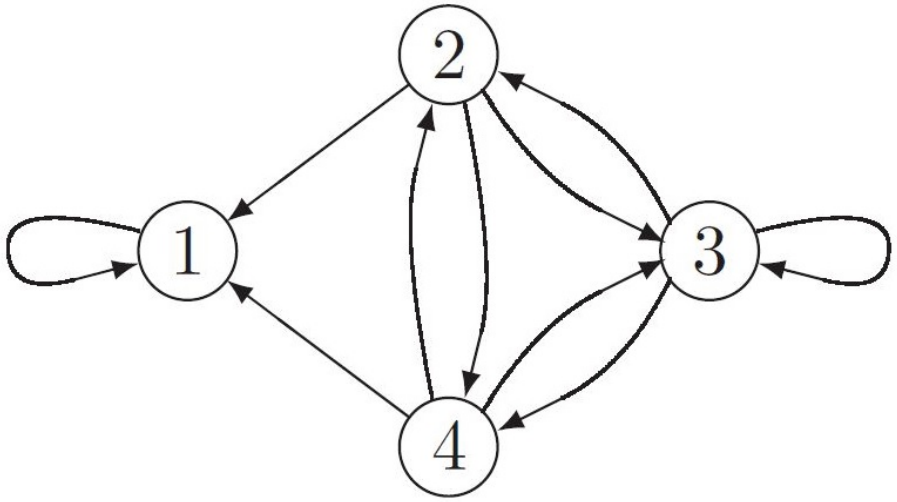
$$r_1 = 0, \quad r_i = 1 \ \text{ for } i \neq 1.$$

Expected First-passage Times

For each sample path starting from state $i \neq 1$, the probability of the initial segment until 1 is entered is unchanged, so the expected first-passage time is unchanged.
The modified Markov chain is now an ergodic unichain with a single recurrent state.

For the chain in Figure (b), we have:

$$\begin{cases} v_2 = 1 + P_{23} v_3 + P_{24} v_4 \\ v_3 = 1 + P_{32} v_2 + P_{33} v_3 + P_{34} v_4 \\ v_4 = 1 + P_{42} v_2 + P_{43} v_3 \end{cases}$$



(b)

Expected First-passage Times

The expected first-passage time to state 1 from state $i \neq 1$ is then

$$v_i = 1 + \sum_j P_{ij} v_j \quad \text{with } v_1 = 0.$$

This can be expressed in vector form as

$$\boldsymbol{v} = \boldsymbol{r} + [P]\boldsymbol{v},$$

where $\boldsymbol{r} = (0, 1, \ldots, 1)^\top$, $v_1 = 0$ and $P_{11} = 1$. Note that if $\boldsymbol{v}$ satisfies $\boldsymbol{v} = \boldsymbol{r} + [P]\boldsymbol{v}$, then $\boldsymbol{v} + \alpha \boldsymbol{e}$ also satisfies it, so that $v_1 = 0$ is necessary to resolve the ambiguity.
Also, since $[P]$ has 1 as a simple eigenvalue, this equation has a unique solution with $v_1 = 0$.

Expected First-passage Times

Example 1: Finding a string in a binary sequence

Assume a Bernoulli process $\{X_i : i \geqslant 0\}$

$$\Pr\{X_i = 1\} = p_1 \quad \text{and} \quad \Pr\{X_i = 0\} = p_0 = 1 - p_1.$$

A binary string $\underline{a} = (a_1, \ldots, a_k)$ is said to occur at time n if

$$(X_{n-k+1}, \ldots, X_n) = (a_1, \ldots, a_k).$$

We want to find the expected time n until this first occurrence.

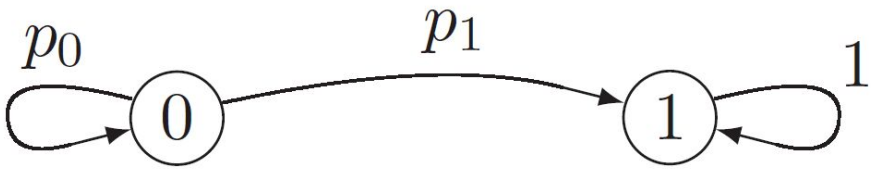
Expected First-passage Times

- The string is a single binary digit, say 1.

  Then it occurs at time n with probability $p_0^{n-1}p_1$. The expect value is then

$$\sum_{n=1}^{\infty} np_0^{n-1}p_1 = \frac{1}{p_1}.$$

  Posed as an expected first-passage time problem:



$$v_0 = 1 + p_0 v_0 \qquad \Rightarrow \qquad v_0 = \frac{1}{p_1}.$$

Expected First-passage Times

- A slightly less trivial case occurs with the string $\underline{a} = (1, 0)$.

  Solution 1: The expected time until the first 1 appears is $1/p_1$ as before. There is a subsequent delay of zero or more subsequent 1' s until the next 0 appears. At this point, the string (1, 0) has just occurred and the additional expected delay is $1/p_0$. Thus the overall delay is

$$\frac{1}{p_1} + \frac{1}{p_0} = \frac{1}{p_0 p_1}$$

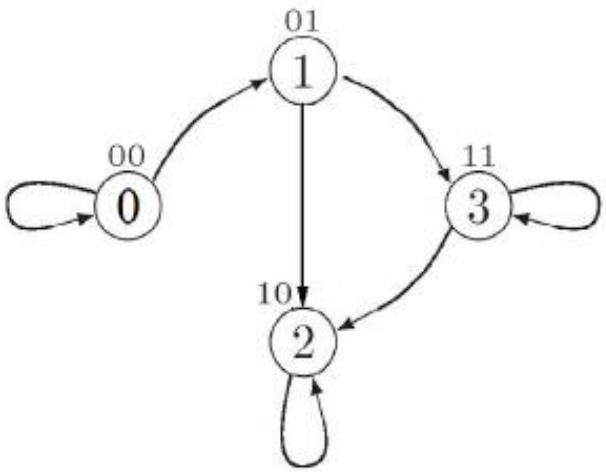
Expected First-passage Times

- A slightly less trivial case occurs with the string $\underline{a} = (1, 0)$.

  Solution 2: Let $Z_n = (X_{n-1}, X_n)$, initializing $Z_0 = (0, 0)$. The expected first-passage time.

$$\begin{cases} v_0 = 1 + p_0 v_0 + p_1 v_1 \\ v_1 = 1 + p_1 v_3 \\ v_3 = 1 + p_1 v_3 \end{cases}.$$

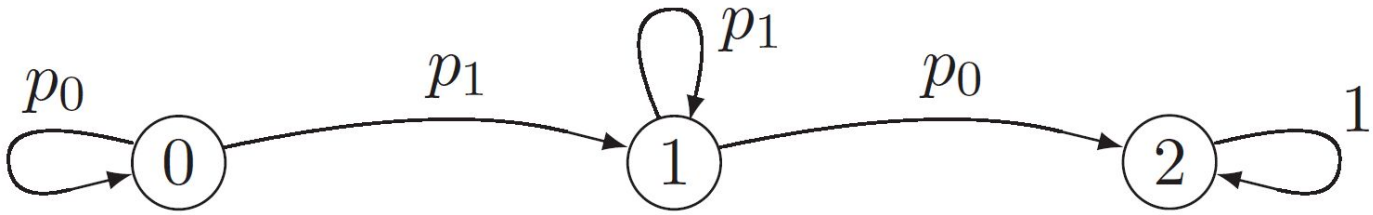
$$v_0 = \frac{1}{p_0 p_1}.$$



Expected First-passage Times

- A slightly less trivial case occurs with the string $\underline{a} = (1, 0)$.

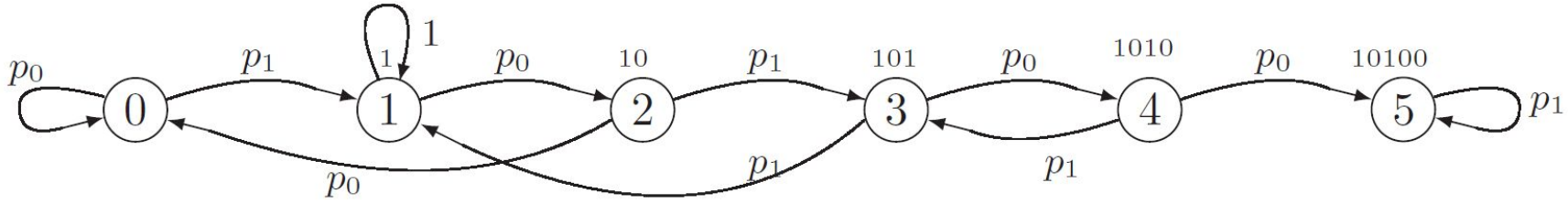
  Solution 3: construct a Markov chain with $k+1$ states. We define the Markov chain to be in state $i < k$ at time n if $(a_1, \ldots, a_k)$ has not yet occurred and if $(X_{n-i+1}, \ldots, X_n)$ is the longest suffix of $X_1, \ldots, X_n$ that forms a proper frefix of $(a_1, \ldots, a_k)$.



$$\begin{cases} v_0 = 1 + p_0 v_0 + p_1 v_1 \\ v_1 = 1 + p_1 v_1 \end{cases} \quad \Rightarrow \quad v_0 = \frac{1}{p_0 p_1}.$$

Expected First-passage Times

- A more general case with the string $\underline{a} = (1, 0, 1, 0, 0)$.



State 4, for example, indicates that the string 10100 will occur if the next bit is 0. If the next bit is 1, then the most recent five bits will be 10101, and the final three of these are the first three bits of 10100, leading to state 3.

4.5.1 Expected First-passage Times

Example 2

Assume that a Markov chain has M states $\{0, \ldots, M-1\}$ and that the state represents the number of customers in an integer-time queueing system.

Suppose we wish to find the expected sum of the customer waiting times, starting with i customers in the system at some given integer time t and ending at the first integer time at which the system becomes idle. For each new customer entering before the system next becomes idle, the waiting time is counted from entry to exit.

We take $r_i = i$ as the 'reward' in state i, and v_i as the expected aggregate reward until the trapping state is entered. Thus

$$v_i = r_i + \sum_{j \geqslant 1} P_{ij} v_j \quad \text{for } i \geqslant 1, \quad \text{and} \quad v_0 = 0.$$

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§ Exercise

Exercise

1. 在瓮中总含有两个球。球的颜色有红色与蓝色。每个时期随机地取出一个球，并且放回一个新球，新球的颜色以 0.8 的概率与取的球同色，而以 0.2 的概率为相反的颜色。如果开始时两个球都是红色，求第五次取到的球是红色的概率。

2. 假定球逐个地被分配到 8 个瓮中，各球以相等的可能放到其中任意一个瓮中。问在分配 9 次后，其中恰有 3 个瓮不是空的概率是多少？

3. 在一系列独立抛掷一个公平硬币的试验中，以 N 记直至出现连续 3 次正面时的抛掷次数。求

- $P(N \leq 8)$
- $P(N = 8)$

Exercise

4. 考虑 4 状态 Markov 链，转移概率矩阵为

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/4 & 1/4 & 0 \\ 1/3 & 0 & 1/3 & 1/3 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1/6 & 1/3 & 1/2 \end{pmatrix}$$

计算从各状态出发到状态 1 的平均首次到达时间，及从状态 1 出发到各状态的平均首次到达时间。

5. 甲，乙，丙三人玩游戏。试讨论规则的公平性：丙是仲裁，掷一均匀硬币多次。

1. 若第一次出现正面，则甲胜；否则，乙胜。
2. 若前三次出现的是“正正反”，则甲胜；若前三次出现的是“正正正”，则乙胜；其他情形，和。
3. 依次记录掷硬币结果，若“正正反”先出现，则甲胜；若“正正正”先出现则乙胜。
4. 依次记录掷硬币结果，若“正正反”先出现，则甲胜；若“正反反”先出现，则乙胜。

Thank you for your attention!